

Total No. of Questions : 8]

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Roll No

CS-228-CBCS

B.E. IV Semester

Examination, June 2020

Choice Based Credit System (CBCS) Analysis and Design of Algorithm

Time : Three Hours

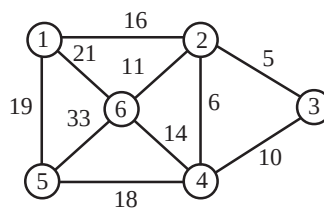
Maximum Marks : 60

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume any data suitably.

1. a) Explain Strassen's algorithm for matrix multiplication with an example.
b) Illustrate Heap Sort in the array
 $A = [5, 8, 3, 9, 2, 10, 1, 40]$
2. a) Explain job sequencing problem with example.
b) Obtain a set of optimal Huffman codes for the seven messages ($M_1, \dots, M_7$) with relative frequencies ($q_1, \dots, q_7$) = (4, 5, 7, 8, 10, 22, 15). Draw the decode tree for this set of codes.
3. a) Consider the Knapsack instance $n = 3$, $(w_1, w_2, w_3) = (2, 3, 4)$ and $(p_1, p_2, p_3) = (1, 2, 5)$ and $m = 5$. Find the optimal solution.
b) Find the shortest path for vertex ① to vertex ③ in the following graph using Dijkstra's greedy algorithm.



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4.
 - a) What is multistage graph problem? Discuss its solution based on dynamic programming approach. Also give a suitable algorithm and find its computing time?
 - b) Explain Hamiltonian cycle and write an algorithm to find all Hamiltonian cycle in graph.
5.
 - a) Explain Dynamic Programming concept with example.
 - b) What is principle of optimality. Give suitable example.
6.
 - a) What is branch and bound techniques?
 - b) Create a B-tree for the following list of elements. (86, 50, 40, 3, 94, 10, 70, 90, 110, 113, 116). Give minimization factor $t = 3$, min degree = 2, max degree = 5.
7.
 - a) Explain the difference between Big - Oh (O), Omega (ω) and theta (θ) notation.
 - b) Explain 2-3 tree with the help of suitable example.
8. Write short notes on:
 - a) Parallel Algorithm
 - b) NP completeness
 - c) Reliability Design
 - d) Merge Sort
